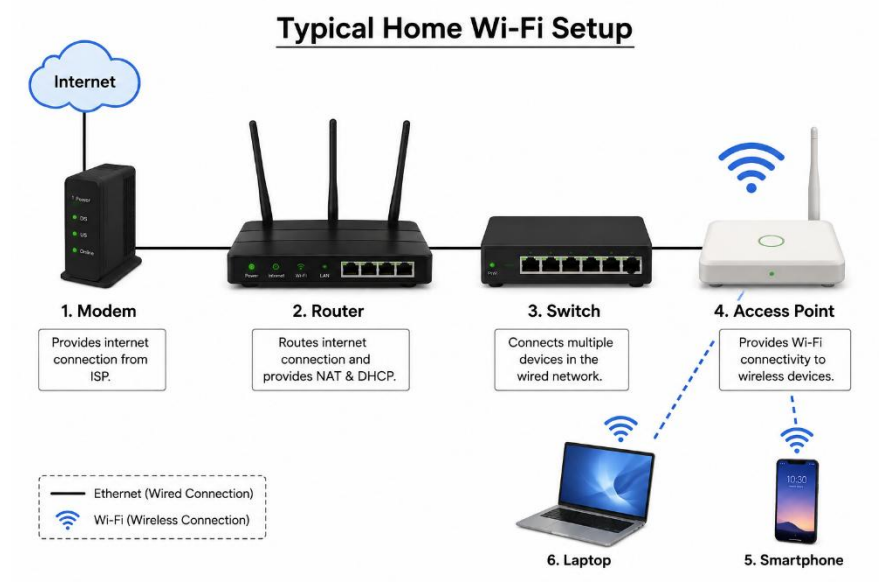


ASSIGNMENT:1

Introduction to Networking Hardware Theory

- ❖ **Task (1): Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.**

Ans.



- ❖ **Task (2):** List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Ans.

- **Networking Device:-**

1.Modem:

- **Main Function:-** Connects your home network to the Internet from the ISP.
- **Real-Life Example:-** Used at home to receive internet from the internet provider.

2.Router:

- **Main Function:-** Shares the internet connection and directs data between devices.
- **Real-Life Example:-** Used at home to share Wi-Fi with phones and laptops.

3. Switch:

- **Main Function:-** Connects multiple wired devices in the same network.
- **Real-Life Example:-** Used in an office to connect computers and printers.

4. Access Point:

- **Main Function:-** Provides wireless Wi-Fi connection to devices.
- **Real-Life Example:-** Used in schools or offices to provide Wi-Fi.

5. Network Cable:

- **Main Function:-** Carries data between networking devices.
- **Real-Life Example:-** Ethernet cable used to connect a PC to a router.

- ❖ **Task (3):** Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Ans.

1. Ethernet Cable (RJ45)



Common use: Used to connect computers, routers, switches, and printers in a wired network.

Source: TP-Link – RJ45 and Ethernet cable

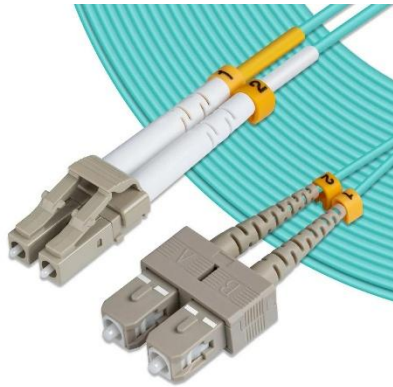
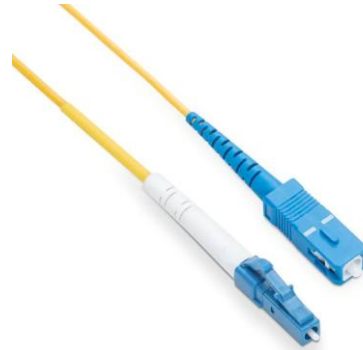
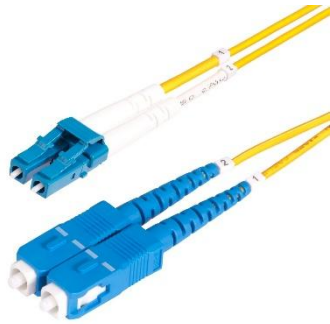
2. Coaxial Cable:



Common use: Commonly used for cable TV, satellite TV, and cable broadband Internet.

Source: TechTarget – Coaxial Cable

3. Fiber Optic Cable:



Common use: Used for high-speed Internet, telecommunications, and long-distance data transmission.

Source: TechTarget – Fiber Optics

- ❖ **Task (4): Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.**

Ans.

Difference	Router	Switch
Main Use	Connects different networks and shares Internet.	Connects multiple devices within the same network.
Typical Location	Usually found at home or office network entrance.	Usually found in offices, labs, or server rooms.
Data Handling	Directs data between different networks using IP addresses.	Sends data between devices using MAC addresses.
Internet Conn.	Usually connects the local network to the Internet.	Normally does not directly provide Internet access.
Example	Home Wi-Fi router connects phones and laptops to the Internet.	Office switch connects computers and printers together.